

How Do Large Vision-Language Models See Text in Image?

Unveiling the Distinctive Role of OCR Heads



Ingeol Baek

LILAB

Department of Artificial
Intelligence
Chung-Ang University

CONTACT.

Email. baekingel98@gmail.com

CV. <https://baekingel.github.io/>

LinkedIn. <https://www.linkedin.com/in/ingeolbaek/>

Scholar. <https://www.linkedin.com/in/ingeolbaek/>

How Do Large Vision-Language Models See Text in Image?

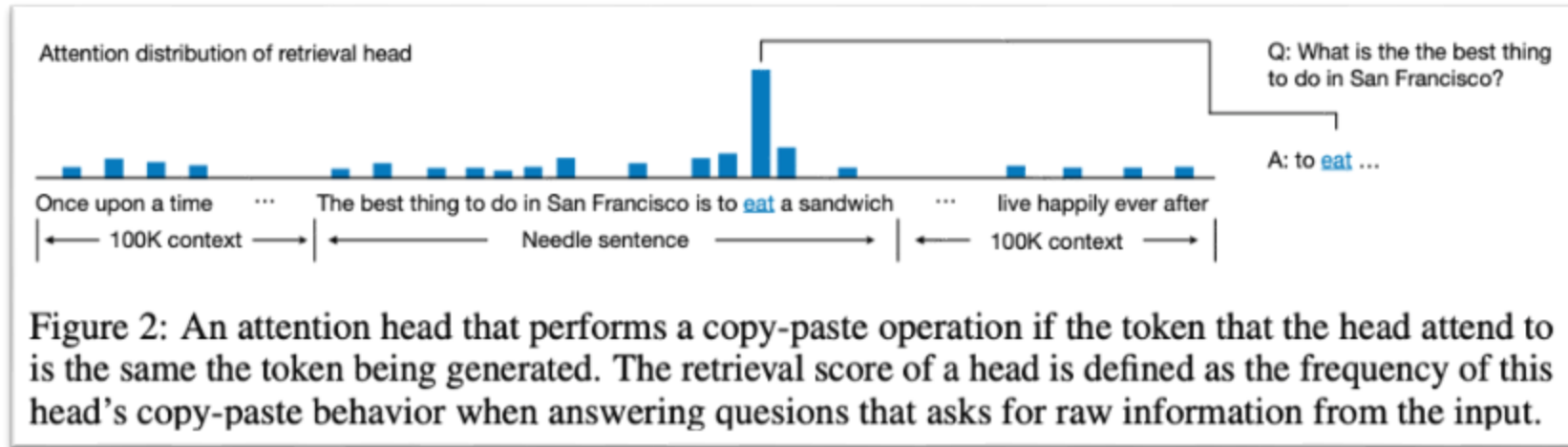
Unveiling the Distinctive Role of OCR Heads – EMNLP 2025

Outline

- Background
- Motivation
- Methods
- Experiment
- Analysis

OCR Head

Background



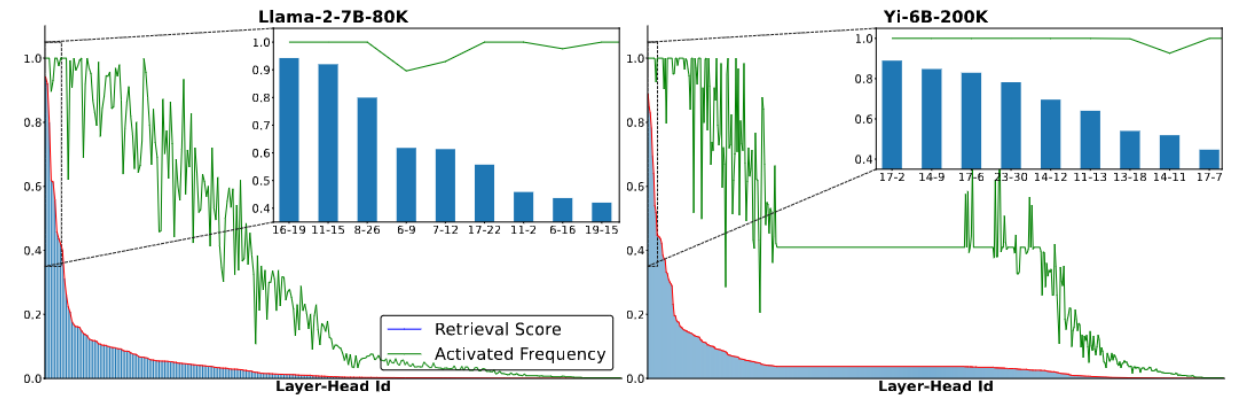
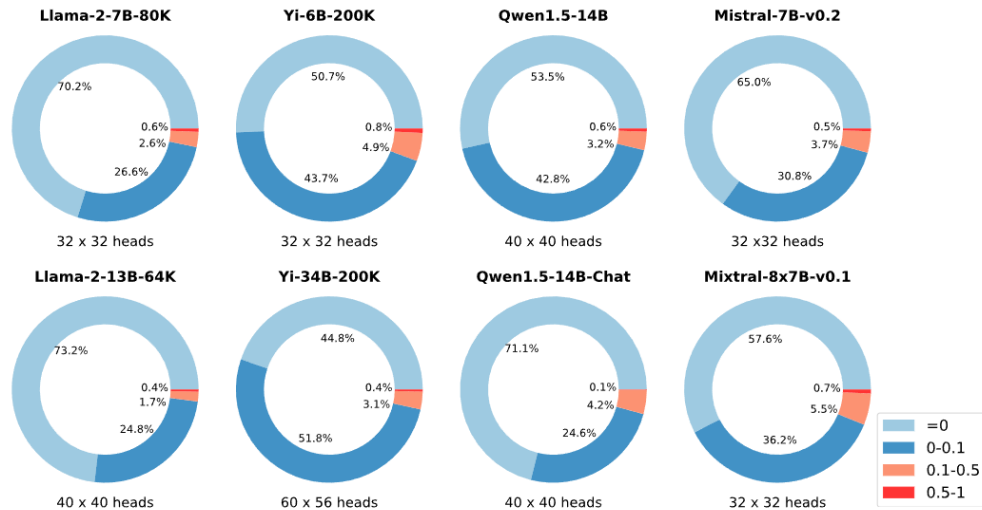
$$\text{Retrieval score for head } h = \frac{|g_h \cap k|}{|k|},$$

Retrieval head:

- We hypothesize that certain attention heads perform **copy-paste** of the correct answer from the context.
- To identify them, we use a **needle-in-a-haystack (NIAH)** dataset that embeds **intermediate answers**.
- While generating the final answer, we assign that head a **Retrieval Score** if, for the gold answer tokens, the head's **maximum attention** falls on the corresponding answer tokens in the context.

OCR Head

Background



- **Retrieval heads** are extremely rare, accounting for only about **0.3–0.5%** of all heads.
- The large gap between their **Retrieval Score** and **Activated Frequency** suggests that these heads are also activated for functions **beyond retrieval**, not exclusively for copying answers.

OCR Head

Motivation



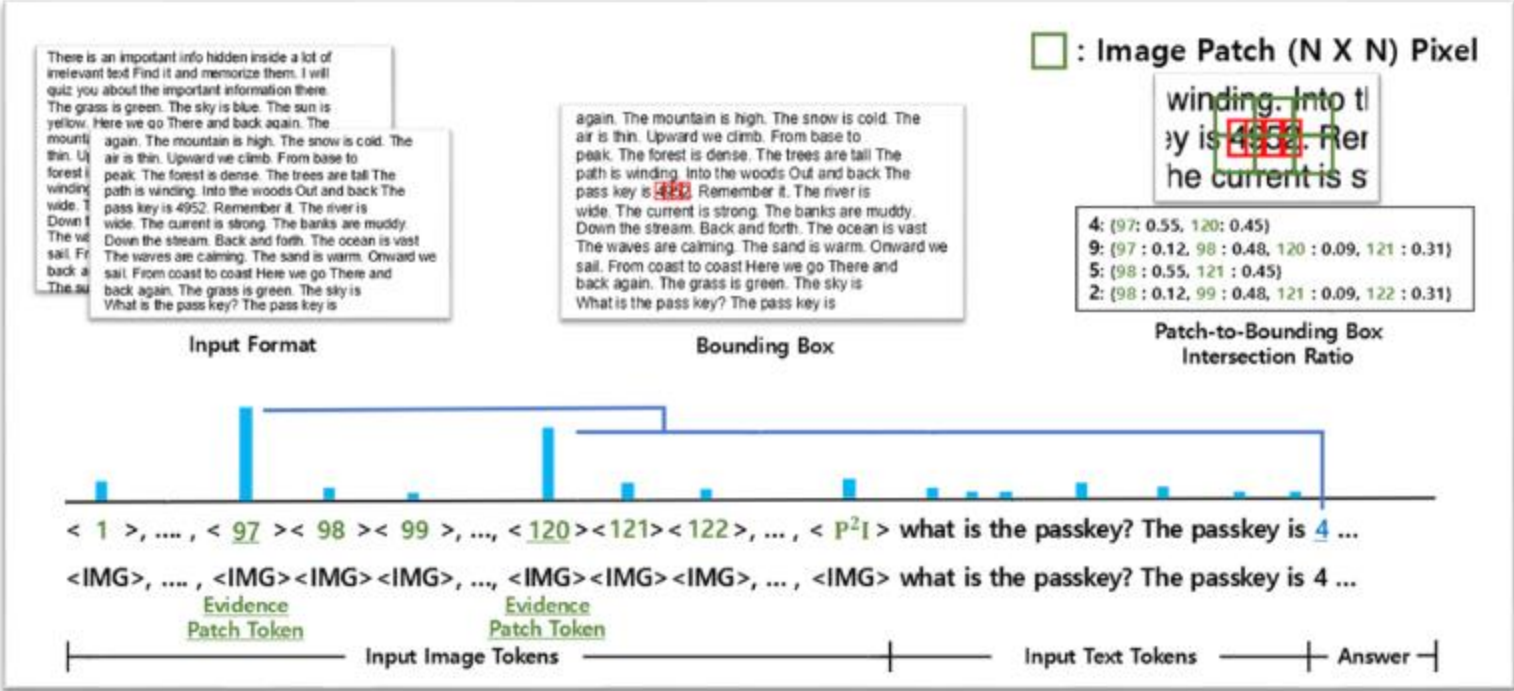
Motivation:

Prior work has shown that in **LLMs**, certain attention heads act as **retrieval heads**, responsible for locating specific content within the in-context document.

This raises a question: in a **Large Vision-Language Model (LVLM)** that combines a ViT, an adaptor, and an LLM, could there exist an attention head specialized in **OCR**?

OCR Head

Method

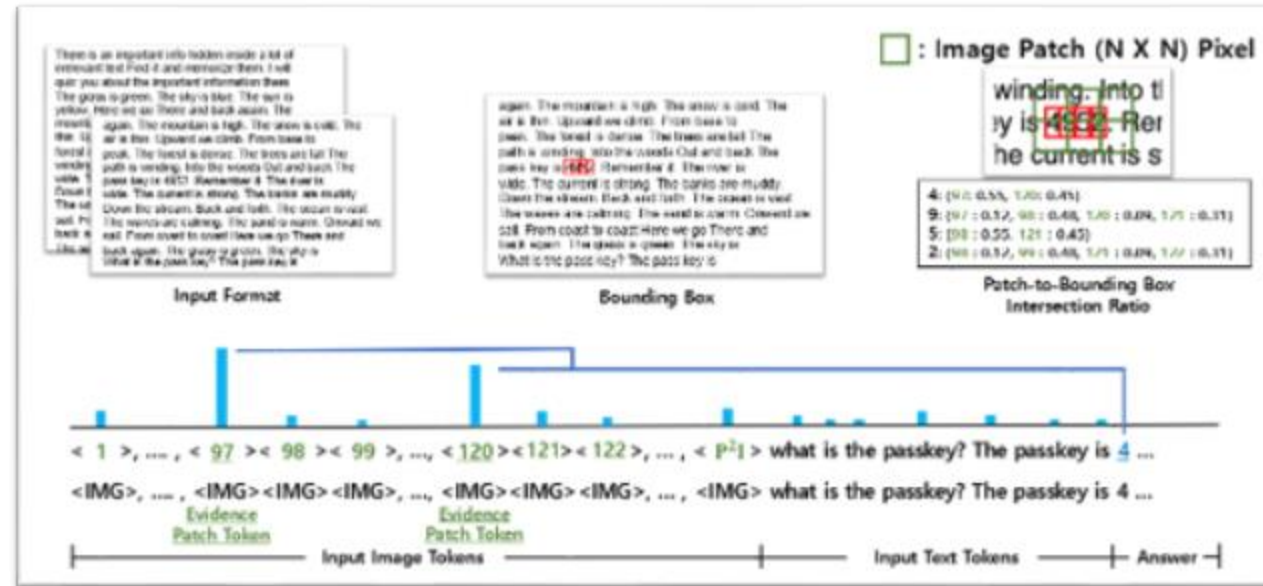


Step1: Image passkey and Needle in a Haystack Dataset Construction

the original text data is converted into text images of a fixed size. Sentences are split whenever they reach a predefined string length, and a new image is generated once the limit is exceeded using a rule-based approach. For the gold answer context, bounding boxes are additionally created to mark the relevant regions.

OCR Head

Method



Step2: Patch Token Preprocessing

assuming a ViT patch size of **14×14**, an image of size **294×196** produces

$$294 / 14 \times 196 / 14 = 294$$

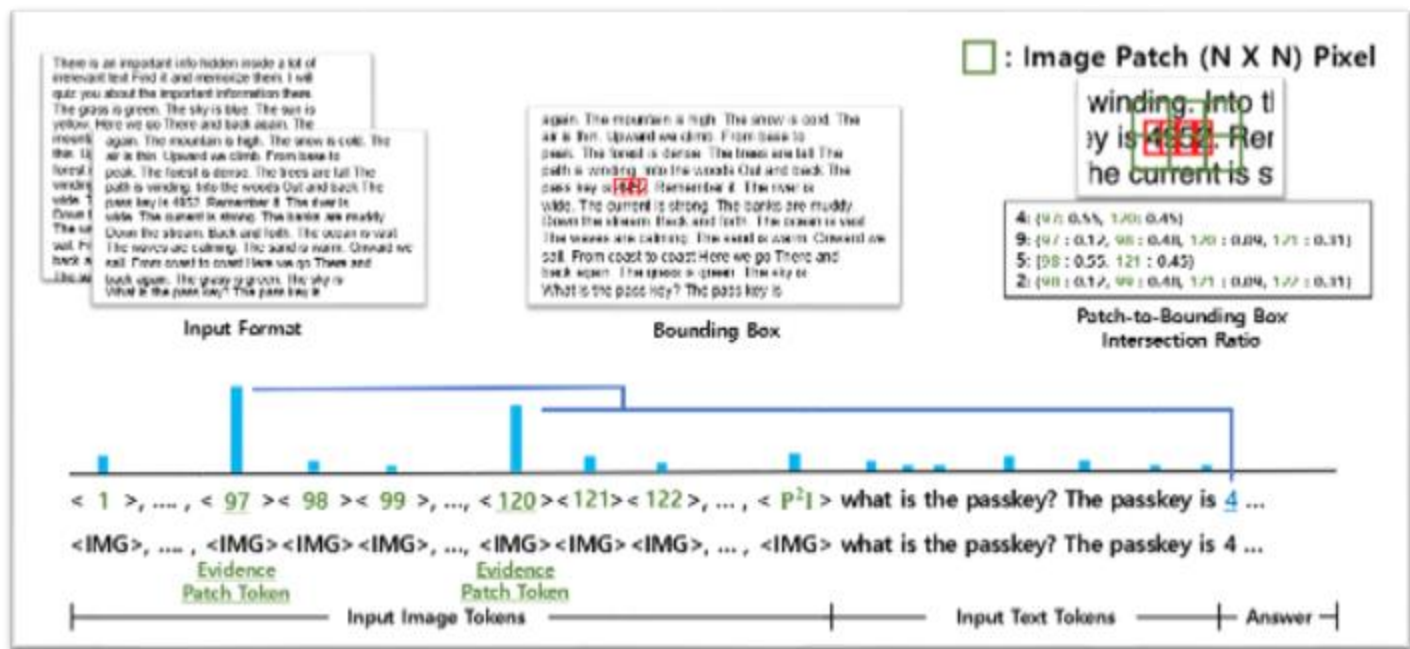
tokens that serve as input to the LLM. When multiple images are considered, the total number of input tokens is calculated as

$$(w / N) \times (h / N) \times I,$$

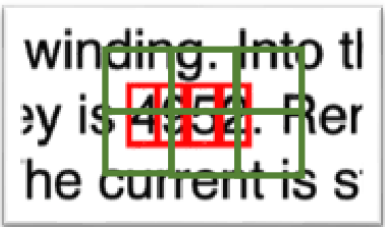
where w and h denote the image width and height, N the patch size, and I the number of images.

OCR Head

Method



□ : Image Patch (N X N) Pixel



4:	{97: 0.55, 120: 0.45}
9:	{97: 0.12, 98: 0.48, 120: 0.09, 121: 0.31}
5:	{98: 0.55, 121: 0.45}
2:	{98: 0.12, 99: 0.48, 121: 0.09, 122: 0.31}

Patch-to-Bounding Box Intersection Ratio

Step3: Evidence Patch Tokens

The process for creating the **Patch-to-Bounding Box Intersection Ratio** shown above involves checking how much each patch overlaps with the bounding box.

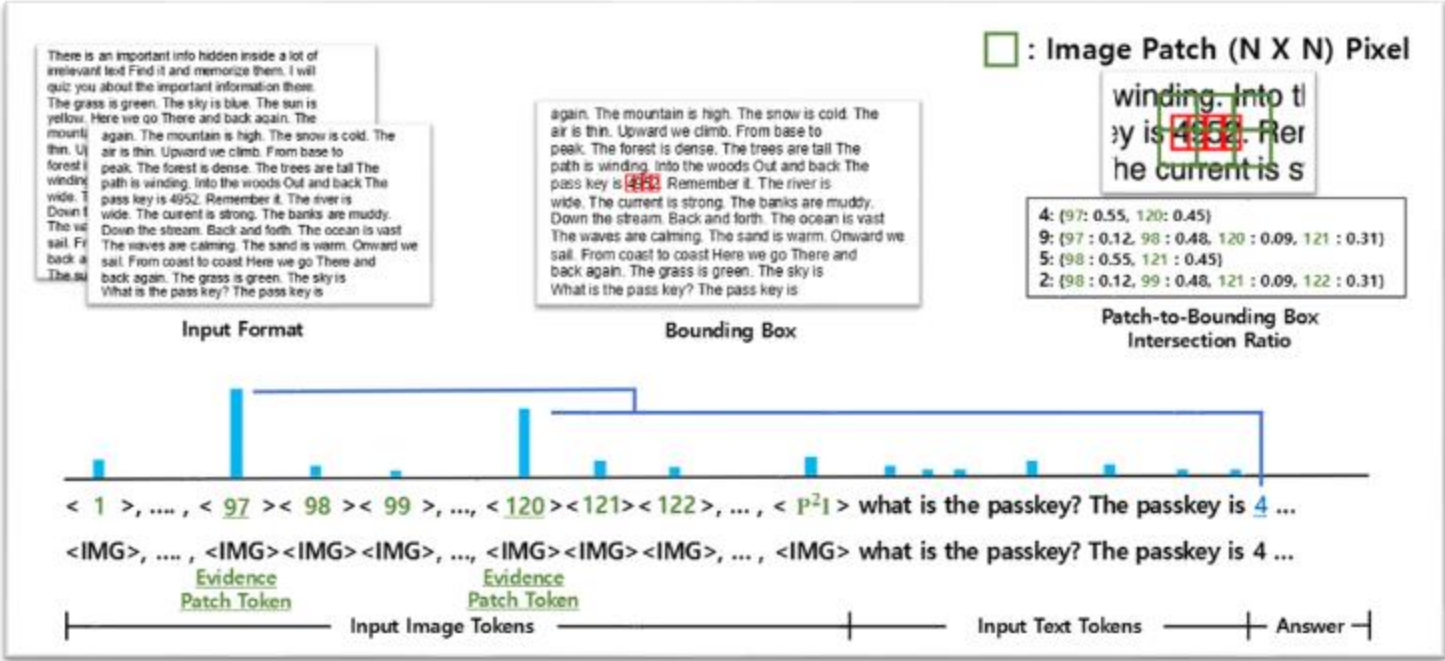
For example, in the case of the gold answer digit “4”, the 97th patch token overlaps by **55%**, while the 120th patch token overlaps by **45%**.

Patches with less than **10% overlap** are discarded.

OCR Head

Method

$$\text{OCR Score for head } h = \frac{|g_h \cap k|}{|k|}$$



Step4: OCR Score

The generated token is considered correct if its **maximum attention score** points to a token within the **Patch-to-Bounding Box Intersection Ratio**.

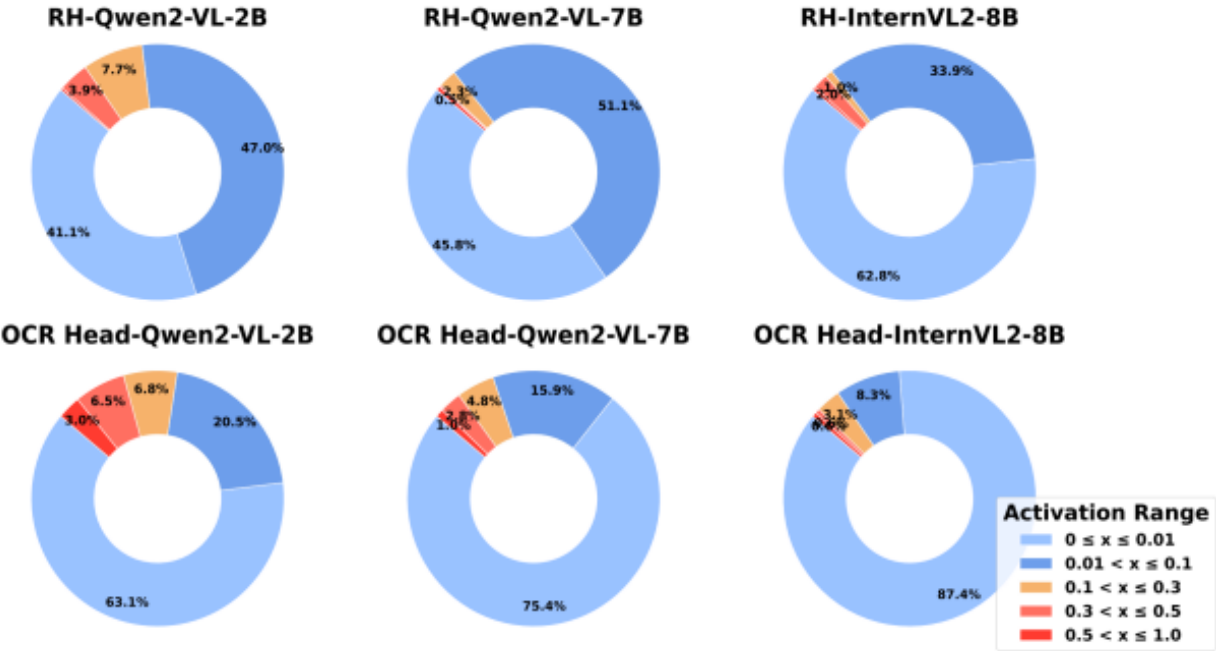
In this case, the generated token corresponds to the **gold answer token**.

Step5: Detecting OCR Head

Using **600 synthetic Image Passkey samples** and **600 Needle-in-a-Haystack samples**, we classify a head as an **OCR head** if it is activated more than **10% on average** and achieves an **OCR Score above 0.1**.

OCR Head

Experiment



Model	Avg Retrieval Score	Similarity
Qwen2-VL-7B Instruction	0	0.507
	0-0.1	0.196
	0.1-0.3	0.104
	0.3-0.5	0.083
	0.5-1.0	0.076
	0.1 ≤	0.347
Model	Avg Retrieval Score	Similarity
InternVL2-8B Instruction	0	0.593
	0-0.1	0.093
	0.1-0.3	0.000
	0.3-0.5	0.000
	0.5-1.0	0.000
	0.1 ≤	0.133

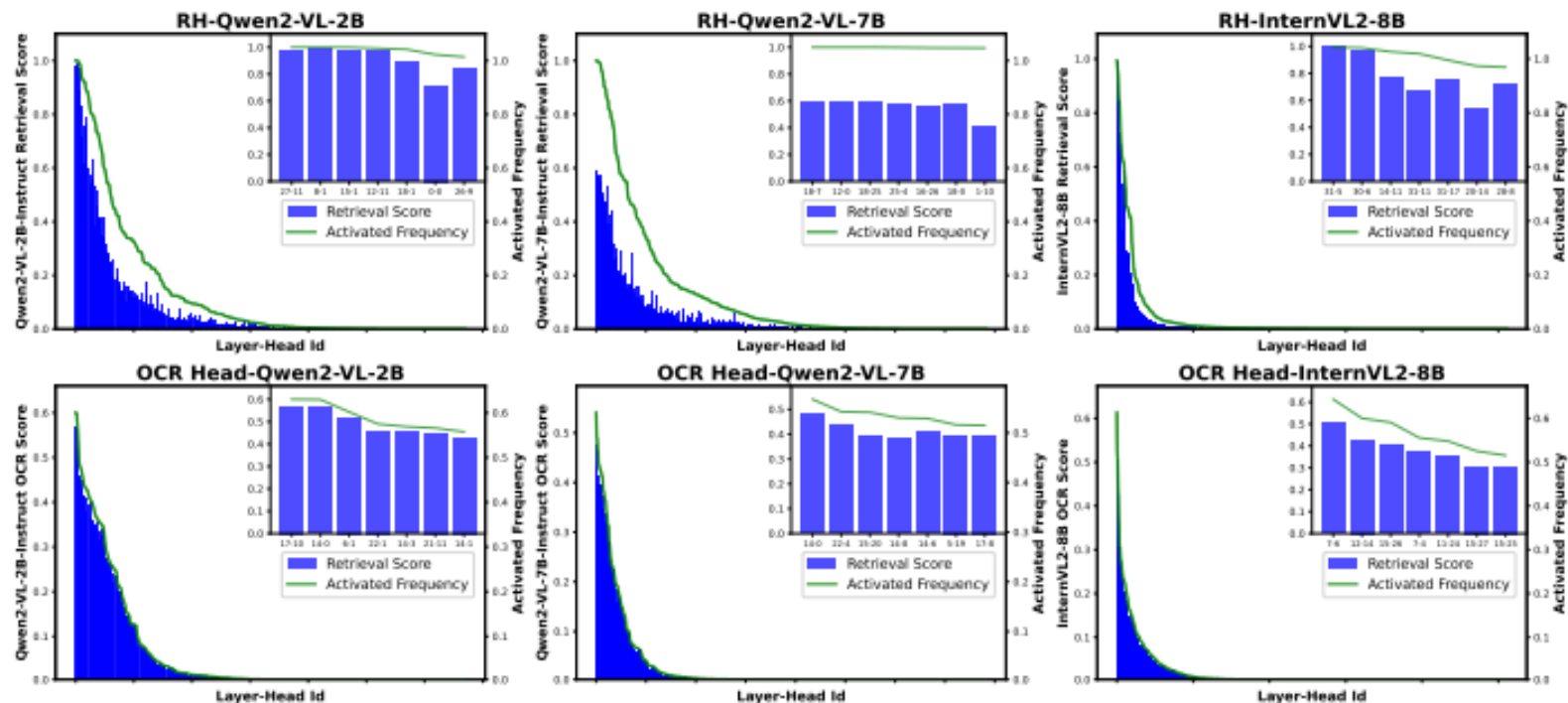
Prior work notes that **Retrieval Heads** are activated **sparsely**.
In contrast, **OCR Heads** show **lower sparsity**, meaning they activate more frequently.

Given their copy-paste-like behavior, similar to Retrieval Heads, we measured **head-to-head similarity** using the **Jaccard index**.

The similarity of activated heads turned out to be relatively low: **0.347 in Qwen2-VL** and **0.133 in InternVL2**.

OCR Head

Experiment



Retrieval Heads show a large gap between **frequency** and **score**, indicating that they are **dynamically activated**. In contrast, **OCR Heads** show almost no such gap, suggesting that they are **statically activated**.

OCR Head

Analysis

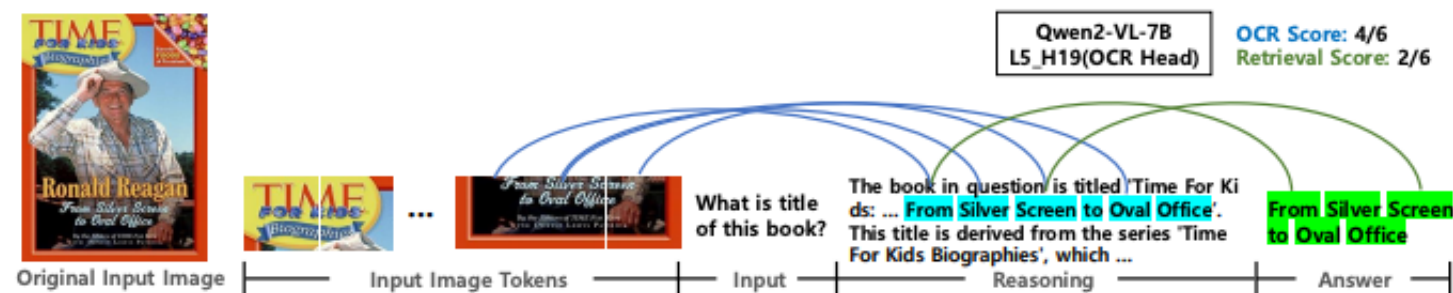


Figure 5: In the OCR-VQA task, we calculate the OCR score and Retrieval score based on the CoT prompting. L5_H19 refers to Layer 5, Head 19. The blue and green lines indicate the token positions with the highest attention.

Qwen2-VL-7B	Method	Score
Top10 OCR Head	Avg OCR Score	0.4004
	Avg Retrieval Score	0.0123
Top10 Retrieval Head	Avg OCR Score	0.2929
	Avg Retrieval Score	0.4705

In the **CoT setting**, we measure both the **OCR Score** (from OCR heads) and the **Retrieval Score** (from retrieval heads), based on (i) the part where the answer is mentioned during reasoning and (ii) the part where the answer is actually generated.

We find that in **OCR heads**, the retrieval score is almost absent, supporting the claim that they are **statically activated**.

In contrast, in **retrieval heads**, an OCR score is also observed, indicating that they are **dynamically activated**.

OCR Head

Analysis

Masked Heads	DocVQA			MP-DocVQA			NQ			HotpotQA		
Qwen2-VL-7B	OCR	RH	Random	OCR	RH	Random	OCR	RH	Random	OCR	RH	Random
Baseline	76.3			69.1			58.0			49.6		
5	74.1	76.1	75.2 ± 0.3	68.1	69.4	67.9 ± 0.2	57.7	57.2	57.6 ± 0.6	51.2	47.8	49.5 ± 0.1
10	73.9	74.6	74.2 ± 0.2	66.7	67.6	66.6 ± 0.4	59.0	53.8	56.4 ± 0.2	49.7	45.3	47.9 ± 1.2
20	59.7	67.1	75.1 ± 0.4	47.7	62.4	67.7 ± 1.7	56.8	36.4	57.3 ± 0.5	47.5	37.3	48.9 ± 0.4
Intern VL2-8B	OCR	RH	Random	OCR	RH	Random	OCR	RH	Random	OCR	RH	Random
Baseline	39.1			35.0			58.3			35.8		
5	36.6	39.7	39.2 ± 0.2	32.8	35.5	35.1 ± 0.6	60.3	56.3	55.9 ± 0.3	33.5	32.8	34.6 ± 0.1
10	30.0	39.7	38.8 ± 0.2	25.1	35.7	34.7 ± 0.4	59.6	55.8	57.8 ± 0.3	33.5	28.2	29.2 ± 0.3
20	24.2	38.2	38.3 ± 0.6	20.7	36.1	34.6 ± 0.3	56.6	58.9	57.6 ± 0.2	34.8	25.7	33.5 ± 1.2

		DocVQA	MP-DocVQA
Qwen2-VL-7B	Baseline	76.3	69.1
	OCR	76.6 (+0.3)	69.7 (+0.6)
	RH	76.2 (-0.1)	69.4 (+0.3)
InternVL2-8B	Baseline	39.1	35.0
	OCR	40.0 (+0.9)	35.8 (+0.8)
	RH	39.3 (+0.2)	34.7 (-0.3)

We conducted an experiment where the **most highly activated heads** in both OCR heads and retrieval heads were **masked**.

Masking OCR heads led to a **significant performance drop** in the **VQA task**.

Masking retrieval heads caused a **large performance decline** in the **QA task**.

To further analyze, we applied **sink token redistribution**.

The sink token is defined as the **first token (index 0)**, and its attention weight was reduced to redistribute scores across other tokens.

When redistribution was applied to OCR heads, we observed a **performance improvement** in the **VQA task**.

OCR Head

Analysis

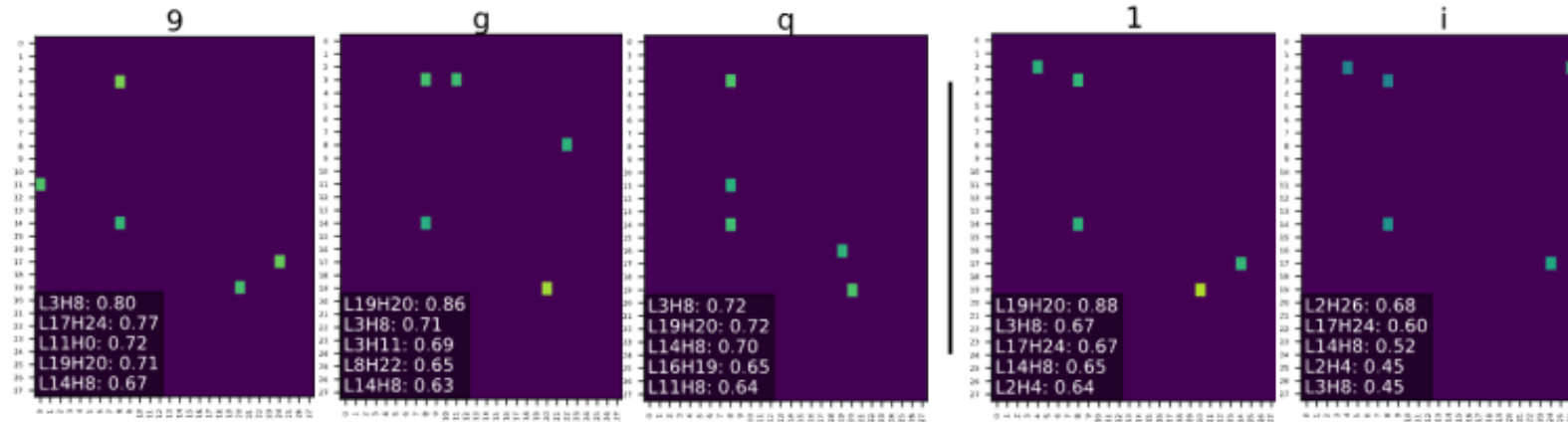


Figure 6: Character-wise activation of top-5 OCR heads; visually similar shapes co-activate common heads.

As the final **case study**, we modified the answer portion of the passkey data into **single digits or single-character answers**, and measured the **average OCR score** across OCR heads.

When observing the top-5 activated heads:

For visually similar characters such as “i” and “1”, the same four heads (**L3H8**, **L2H4**, **L14H8**, and **L17H24**) were activated.

For “9”, “g”, and “q”, three heads were commonly activated across these characters.

Thank you for listening